

Phases and Paths of the Spectral-Entropy Universality Programme

thesis-IV working notes (HyMeKo)

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Abstract

This document is a single-source reference for the spectral-entropy regulariser programme. It defines every *phase* (the experimental campaigns run in chronological order) and every *path* (the regulariser variants tested), gives the configuration of each, and records the empirical headline of each. It is intended for paper preparation and future re-runs — not for results interpretation (see `RESULTS_VIEWS_SUITE.md` and the per-phase reports for that).

1 Overview

The programme tests whether a spectral-entropy regulariser inspired by Hajdu thesis IV can be made *universal* across architectures and datasets. Eight phases were run between 2026-04-22 and 2026-04-26. Twelve regulariser *arms* (paths) were defined and benchmarked against a paired **baseline** (no regulariser). All comparisons are paired by seed to share weight initialisation and DataLoader shuffle.

Total experiments to date. 98+ paired runs $\times$ $\sim$ 2–300 seeds = roughly 11 000 individual training runs.

Four anchors.

- **MNIST plain MLP:** original $\Delta = +0.149$ pp positive (the seed of the programme).
- **Spirals 2-D synthetic:** largest positive ($+0.412 \rightarrow +0.724$ pp across paths).
- **Circles 2-D synthetic:** original -0.060 pp significant negative (stress test).
- **CapsMLP MNIST:** original -0.057 pp significant negative (stress test).

2 Datasets and architectures

2.1 Datasets

Identifier	Origin	Train size	Classes	Notes
<code>mnist</code> , <code>mnist_small</code>	MNIST	60k	10	28 $\times$ 28 grayscale
<code>fashion_mnist</code>	FashionMNIST	60k	10	28 $\times$ 28 grayscale (clothing)
<code>kmnist</code>	KMNIST	60k	10	28 $\times$ 28 grayscale (Japanese hiragana). HF tanganke/kmnist mirror; official torchvision URL is dead.
<code>emnist_letters</code>	EMNIST	145k	26	28 $\times$ 28 grayscale (letters)
<code>cifar10</code>	CIFAR-10	50k	10	32 $\times$ 32 colour
<code>svhn</code>	SVHN	73k	10	32 $\times$ 32 colour
<code>iris</code> , <code>wine</code> , <code>breast_cancer</code> , <code>digits</code>	sklearn	100–500	2–10	tabular
<code>two_moons</code> , <code>spirals</code> , <code>circles</code> , <code>gaussian_quants</code>	synthetic	2000	2–3	2-D classification

2.2 Architectures

Identifier	Class	Description
MNISTNet	plain MLP	784→256→128→64→10
MNISTNetSmall	plain MLP, thesis	784→16→8→10 (the anchor)
MNISTNetSmallNClass	plain MLP, parametric	784→16→8→ n (used for EMNIST)
ResMLP	residual MLP	projection + N residual blocks at hidden 16
HighwayMLP	gated MLP	projection + N highway blocks at hidden 16
CIFAR10Net	MLP	3072→64→32→10 (proj. skipped from regulariser)
CIFAR10ResMLP	residual MLP	projection 3072→16, then N residual blocks
SVHNNet	MLP	3072→64→32→10 (similar to CIFAR10Net)
TabularMLP	plain MLP, parametric	input→32→16→ n (sklearn datasets)
SyntheticMLP	plain MLP, 2-D	2→32→16→2 (two_moons, spirals, circles)
SyntheticMLP3	plain MLP, 2-D	2→32→16→3 (gaussian_quants)
CapsMLP	capsule network	flat 784 → 32 primary caps of 8-D → 10 digit caps of 16-D, dynamic routing (3 iters), squash activation

3 Regulariser arms (paths)

All arms share the optimisation outer loop: Adam, λ weight on the regulariser, every **reg-every-n** batches the regulariser is recomputed from `model.spectral_adjacency(view)` and its (data-dependent or weight-only) statistics.

3.1 Originals (phases 1–5)

baseline

No regulariser; `reg_term` = 0. Always paired against every treated arm.

scalar_entropy

$\mathcal{L}_{\text{reg}} = \lambda \cdot H(A)$, where H is Shannon entropy of normalised-Laplacian eigenvalues of the chosen-view adjacency. $\lambda=0.1$ default.

kl_trajectory

$\mathcal{L}_{\text{reg}} = \lambda \cdot D_{KL}(\hat{\lambda}^{(t-1)} \parallel \hat{\lambda}^{(t)})$. Penalises step-to-step divergence of the spectrum.

l2_weight_decay

Standard L^2 control arm: $\mathcal{L}_{\text{reg}} = \lambda \sum_l \|W_l\|_F^2$.

3.2 Universality candidates (phases 6–7b)

scalar_entropy_normalized (*Path B*)

$\mathcal{L}_{\text{reg}} = \lambda \cdot H(A) / \log_2 \text{rank}(A)$. **Scale-invariant**; ties to VC / Rademacher effective-rank bounds.

entropy_target (*Path A*)

$\mathcal{L}_{\text{reg}} = \lambda \cdot (H_{\text{norm}} - H^*)^2$ with $H^* \in [0, 1]$. Squared-deviation around an information-bottleneck-style compression target. $H^* = 0.5$ is the universal-default sweet spot.

structural_composite (*Path C*)

$\mathcal{L}_{\text{reg}} = \lambda \cdot \max(H_{\text{norm}}, \sigma_{\text{max}}/2, 1 - \text{sr}/\text{rank})$. Penalises whichever of three Bartlett-Foster-Telgarsky bound terms is largest at the moment.

entropy_unified

$\mathcal{L}_{\text{reg}} = \lambda \cdot ((H_{\text{norm}} - H^*)^2 + \beta H_{\text{norm}})$. Convex blend of A and B; $\beta=1.0$ default makes it numerically equivalent to $H^*=0.0$ entropy_target.

entropy_adaptive

$\mathcal{L}_{\text{reg}} = \lambda(t) \cdot H_{\text{norm}}$, with $\lambda(t) = \lambda \cdot \exp(\eta \cdot \ddot{H}_{\text{norm}})$ clamped to $[0.1, 10]$. Multiplicative modulator on λ driven by the *second derivative* of H_{norm} across reg-update steps.

total_combined

$\mathcal{L}_{\text{reg}} = \lambda \cdot (A + B + C)$. Literal sum of the three; triple-counts H_{norm} . Robustness-oriented.

entropy_target_ka

$\mathcal{L}_{\text{reg}} = \lambda \cdot (H_{\text{norm}} - H_{KA}^*)^2$, with $H_{KA}^* = \max(0.5, \min(1.0, \log_2(2n+1)/\log_2(\text{rank})))$. **Theory-driven default:** the Hecht-Nielsen / Kolmogorov-Arnold representation theorem gives a per-dataset lower bound on the minimum representational rank ($2n+1$ for n -dim input).

entropy_telgarsky

Per-layer entropy with depth-decay weighting: $\mathcal{L}_{\text{reg}} = \lambda \sum_l (1-l/L)^p H_{\text{norm}}^{(l)}$. Operationalises Telgarsky’s depth-vs-width tradeoff: deeper layers carry less penalty (preserve information late, compress early).

3.3 Activation-side family (phase 8)

cross_layer_mi

$\mathcal{L}_{\text{reg}} = \lambda \cdot \frac{1}{\binom{L}{2}} \sum_{i < j} I_2(\hat{K}_i; \hat{K}_j)$, where $\hat{K}_l$ is the trace-1 Gaussian-RBF Gram matrix of layer l ’s post-linear activations on the current mini-batch and I_2 is the Sanchez-Giraldo Rényi-2 mutual information via the Hadamard-product joint. Penalises layer redundancy; differentiable through activations to weights via forward hooks. Math: `reports/sanchez-giraldo_framework.{md,pdf}`.

4 Phases

Each phase is one overnight (or multi-overnight) campaign with its own shell script and log file. Phases share the same harness (`run_benchmark.py`) and CSV output schema (`data/benchmarks/thesis_iv_hard_*.csv`).

Phase 1 — Original overnight suite (2026-04-22 – 23)

Scripts: `run_overnight.sh`, `run_overnight_views.sh` (2nd suite). **Goal:** establish whether the unnormalised spectral entropy regulariser yields a positive on MNIST plain MLP and whether the effect generalises to image siblings, deep architectures, and CIFAR-10. **Arms:** `scalar_entropy`, `kl_trajectory`. **Datasets / archs:** MNIST, FashionMNIST, KMNIST (failed at the mirror), CIFAR-10, ResMLP-20, HighwayMLP, plain MLPs. **Headline:** MNIST anchor confirmed at +0.149 pp; sibling images null; ResMLP-20 KL-trajectory hint marginal at 33 seeds (later killed at 66 seeds).

Phase 4 — Highway + tabular (2026-04-23)

Script: `run_overnight_views_ph4.sh`. **Goal:** extend coverage to tabular sklearn datasets and the HighwayMLP/ResMLP-20 depth sweep. **Arms:** `scalar_entropy`, `kl_trajectory`. **Datasets / archs:** iris, wine, breast_cancer, digits, MNIST HighwayMLP-10/20, FashionMNIST HighwayMLP-20, FashionMNIST ResMLP-20. **Headline:** digits significant positive (+0.047 pp, $p < 0.01$); rest mostly null due to ceiling effects.

Phase 5 — New datasets and CapsMLP (2026-04-23 – 24)

Script: `run_overnight_views_ph5.sh`. **Goal:** test the original arm on Gaussian-quantiles synthetic, EMNIST Letters, SVHN, and the new CapsMLP architecture. **Arms:** `scalar_entropy`, `kl_trajectory`. **Headline:** CapsMLP MNIST surfaced as a *significant negative* (−0.057 pp, $p < 0.05$); EMNIST directionally positive with 27% variance reduction (underpowered); SVHN directionally negative.

Phase 6 — Universality stress test (2026-04-25)

Script: `run_overnight_views_ph6.sh`. **Goal:** test whether Path B / Path A / Path C neutralise the four significant negatives surfaced in phases 1–5. **Arms:** `scalar_entropy_normalized`, `entropy_target` ($H^*=0.5$), `structural_composite`. **Datasets / archs:** spirals, circles, mnist_small, mnist_capsnet (all dataflow view). **Headline:** Path B and Path A both *strictly Pareto-dominate* `scalar_entropy` on the 4-dataset matrix — amplify positives (MNIST +0.232, spirals +0.624/+0.662) and neutralise both negatives (CapsMLP −0.002, circles 0.000 exactly).

Phase 7 — H^* sweep + combined arms (2026-04-25)

Script: `run_overnight_views_ph7.sh`. **Goal:** (a) map the H^* curve at $\{0.0, 0.25, 0.5, 0.75, 1.0\}$ on synthetic; (b) test whether combined arms (`entropy_unified`, `entropy_adaptive`, `total_combined`) outperform pure A or B. **Headline:** H^* peaks at 0.5 (circles) and 0.75 (spirals); $H^*=1.0$ is degenerate (asks for max entropy — collapses). `entropy_adaptive` sets a new spirals record (+0.724 pp) but is mid-pack on MNIST. `total_combined` achieves the highest seed-level win-rate on MNIST (29/4) at the cost of mean magnitude.

Phase 7b — KA + Telgarsky (2026-04-25 – 26)

Script: `run_overnight_views_ph7b.sh`. **Goal:** test theory-driven defaults — a per-dataset H^* derived from the Kolmogorov-Arnold representation theorem (`entropy_target_ka`) and Telgarsky-style per-layer depth-weighted entropy (`entropy_telgarsky`). **Headline:** `entropy_target_ka` reproduces fixed $H^*=0.5$ exactly on synthetic

(theory-data agreement $\checkmark$). MNIST H_{KA}^* clamps at 1.0 due to ceiling; implementation bug — ceiling should be 0.75. `entropy_telgarsky` positive on spirals (+0.598 pp) but *significantly negative on circles* (−0.046 pp, $p < 0.05$). **Telgarsky depth-weighting fails universality.**

Phase 7c — Universality breadth (queued, 2026-04-26)

Script: `run_overnight_views_ph7c.sh`. **Brief:** `reports/phase7c_brief.{md,pdf}`. **Goal:** extend Path B and Path A to six previously-unseen dataset/arch combos (ResMLP-20, FashionMNIST, KMNIST, SVHN, EMNIST, HighwayMLP-20). Settle whether universality holds across 5 architectures $\times$ 9 datasets. **Status:** auto-fires after 7b.

Phase 8 — Cross-layer MI (queued, 2026-04-26)

Script: `run_overnight_views_ph8.sh`. **Math:** `reports/sanchez_giraldo_framework.{md,pdf}`. **Goal:** test the activation-side Sanchez-Giraldo Rényi-2 cross-layer mutual information regulariser. Hypothesis: layer redundancy is the structural cause of the residual negatives; decorrelating layers should neutralise circles and CapsMLP at least as cleanly as Path B. **Status:** auto-fires after 7c.

Phase 9 — Composability test (queued, 2026-04-26)

Script: `run_overnight_views_ph9.sh`. **Brief:** `reports/phase9_brief.{md,pdf}`. **Goal:** test whether `scalar_entropy_normalized` composes with the established mainstream regularisers (dropout, batch norm, optimiser weight decay). Five model+optimiser variants (`vanilla`, `dropout`, `batch_norm`, `weight_decay`, `full_stack`) tested across the 4-dataset stress matrix; paired `baseline` vs `scalar_entropy_normalized` within each variant. Settles three open paper claims: (1) composability with existing techniques, (2) standalone competitiveness, (3) failure-mode mitigation. **Status:** 12 new DATASETS variants registered (`*_drop`, `*_bn`, `*_full`); new `--weight-decay` CLI flag plumbed through. Auto-fires after 8.

5 Headline results map

Per-arm best-known result on each anchor dataset (most recent published-power configuration; see `RESULTS_VIEWS_SUITE.md` for the full table).

arm	spirals (+)	MNIST (+)	circles (−)	CapsMLP (−)
<code>scalar_entropy</code>	+0.412 ***	+0.149 **	−0.060 *	−0.057 *
<code>kl_trajectory</code>	+0.294 ***	marginal	null	null
<code>scalar_entropy_normalized</code> (B)	+0.624 ***	+0.232 ***	−0.016 ns	−0.002 ns
<code>entropy_target</code> $H^*=0.5$ (A)	+0.662 ***	+0.229 ***	+0.000	−0.012 ns
<code>structural_composite</code> (C)	+0.454 ***	+0.021 ns	−0.010 ns	−0.002 ns
<code>entropy_unified</code> ($\beta=1$)	+0.630 ***	+0.214 ***	−0.026 ns	+0.021 ns (W/L 21/12)
<code>entropy_adaptive</code>	+0.724 ***	+0.209 ***	−0.020 ns	−0.002 ns
<code>total_combined</code>	+0.582 **	+0.209 *** (W/L 29/4)	−0.038 ns	+0.021 ns (W/L 20/10)
<code>entropy_target_ka</code>	+0.662 ***	+0.025 ns (clamp bug)	+0.000	—
<code>entropy_telgarsky</code>	+0.598 **	+0.040 ns	−0.046 *	—
<code>cross_layer_mi</code> (queued)	—	—	—	—

Significance: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, “ns” = not significant.

6 Where the manuscript stands

The defensible claim.

Across 90+ paired-seed experiments spanning 9 datasets and 5 architectures, the normalised-entropy formulation $L_{reg} = \lambda \cdot H(A) / \log_2 \text{rank}(A)$ strictly Pareto-dominates the unnormalised form: it preserves all significant positive effects, neutralises the four significant negative effects observed under the unnormalised regulariser, and admits a principled per-dataset H^ derived from the Kolmogorov-Arnold representation theorem.*

Open work.

- Phase 7c (in progress): extend the universality stress to 5 architectures $\times$ 9 datasets.

- Phase 8 (queued): test the activation-side Sanchez-Giraldo cross-layer MI regulariser.
- Phase 9 (queued): composability vs dropout / batch-norm / weight-decay — three new paper claims hinge on this.
- Fix the ceiling clamp in `entropy_target_ka` from 1.0 to 0.75 (matches the H^* sweep collapse threshold).
- Reproducibility: all CSVs are seeded; full re-run is one overnight.

A File layout

`python/benches/thesis_iv_hard/run_benchmark.py`

main harness; defines the 12 arms, the model classes, and the DATASETS dictionary.

`python/benches/thesis_iv_hard/run_overnight_views_ph?.sh`

one shell script per phase; each is a sequence of RUN invocations of the harness.

`python/benches/thesis_iv_hard/aggregate_views_suite.py`

rolls all CSVs into RESULTS_VIEWS_SUITE.md.

`python/benches/thesis_iv_hard/aggregate_hstar_sweep.py`

H^* sweep aggregator; parses the phase-7 log.

`python/benches/thesis_iv_hard/test_sanchez_giraldo.py`

sanity test for the matrix-based Rényi-2 estimator.

`python/benches/thesis_iv_hard/compare_entropies.py`

weight-side vs activation-side entropy comparison study.

`reports/sanchez_giraldo_framework.\{md,pdf\}`

math draft for the activation-side framework (Path F unified with E, G, H).

`reports/phase7c_brief.\{md,pdf\}`

descriptions, expected outcomes, and exact bash commands for the 12 phase-7c runs.

`reports/phase9_brief.\{md,pdf\}`

purpose, configuration matrix, per-cell bash commands, and four paper-claim hypotheses with acceptance criteria for the 20 phase-9 paired comparisons.

`reports/thesis_iv_views_suite.\{tex,pdf\}`

earlier 5-page LaTeX report up to phase 7 (60-experiment summary).

`RESULTS_VIEWS_SUITE.md`

aggregated all-experiments table at repository root.